

To generate a signal with a frequency sweep rate (chirp rate) of 1.9683×10^{13} Hz/sec on a 26 TOPS AI hat, you can use an algorithm that produces a linear ramp control voltage for a voltage-controlled oscillator (VCO). This approach efficiently leverages the AI hat's compute capabilities and available digital-to-analog converter (DAC) to achieve the desired chirp.

Algorithm: Linear Ramp Control for VCO Chirp

Inputs:

- Chirp rate: $\alpha = 1.9683 \times 10^{13}$ Hz/sec
- VCO gain: K_{VCO} (Hz/volt) — assumed known from hardware specs
- Starting frequency: f_0 (Hz) — e.g., 0 Hz or a specified value
- Chirp duration: T (seconds) — e.g., 1 μ s
- DAC sampling rate: F_s (Hz) — must meet Nyquist for the control signal bandwidth

Steps:

1. **Compute control voltage slope:**
 $m = \alpha K_{VCO}$ (volts/sec)
2. **Compute initial control voltage:**
 $V_0 = f_0 K_{VCO}$ (volts)
3. **Determine number of samples:**
 $N = \lceil T \times F_s \rceil$
4. **Generate samples in a loop** (for $n=0$ to $N-1$):
 - Current time: $t = n / F_s$
 - Control voltage: $V = V_0 + m \cdot t$
 - Output V to the DAC (or buffer for later output)

Notes:

- The VCO output frequency is $f(t) = K_{VCO} \cdot V(t)$, resulting in a linear chirp $f(t) = f_0 + \alpha t$.
- Choose F_s high enough to smoothly approximate the ramp (e.g., 40 MHz for a 1 μ s chirp).
- The 26 TOPS AI hat can handle the lightweight computations and real-time DAC updates.
- For longer chirps, ensure the final frequency does not exceed the VCO's range.

Example:

If $K_{VCO} = 1 \times 10^9$ Hz/V, $f_0 = 0$, $T = 1 \mu$ s, and $F_s = 40$ MHz:

- $m = 1.9683 \times 10^{13} / 10^9 = 19683$ V/s
- $V_0 = 0$
- $N = 40$
- Output: 40 voltage samples linearly increasing from 0 to 1.9683×10^{-2} V.

This algorithm is simple, efficient, and suitable for real-time execution on an AI hat with a DAC, enabling precise linear chirp generation for applications like radar or LiDAR.